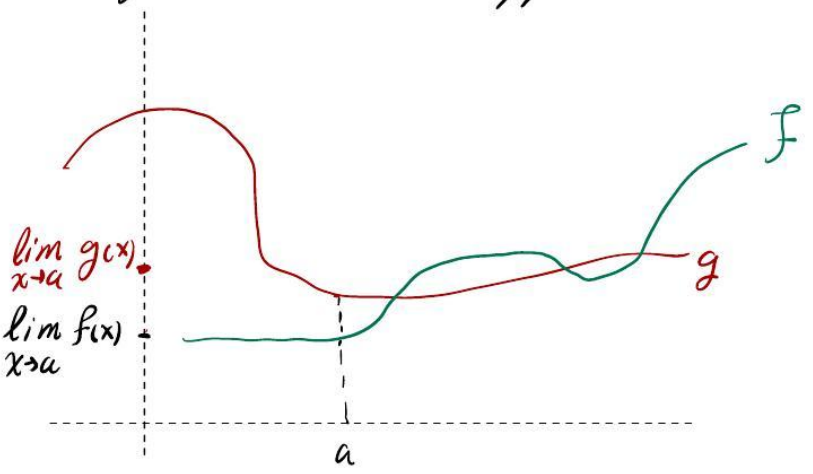
Lecture 4, Dec 22, 2025.

The squeeze theorem. The following theorems describe how the limit of functions are related when the values of one function are greater than those of another.

Thm 1 If $f(x) \le g(x)$ when x is near a and the limits of f and g exist as x approaches a, then.

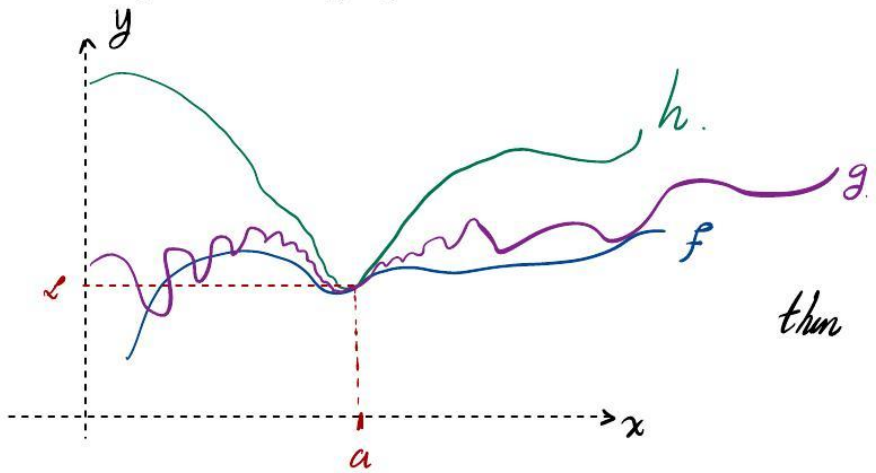


$$\lim_{x\to a} f(x) \le \lim_{x\to a} g(x).$$

Problem. Evaluate $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right)$.

if $f(x) = x^2$ and $g(x) = \cos\left(\frac{1}{x}\right)$ then $\lim_{x\to 0} f(x)\cdot g(x)$. but if you want apply the product law we need to check that both limits exists. However $\lim_{x\to 0} \cos\left(\frac{1}{x}\right)$ = DNE.

The Squeese theorem (Sandwich theorem). If $f(x) \le g(x) \le h(x)$ when x is near a and $\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$.



then $\lim_{x\to a} g(x) = L$.

Example. Show that $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$.

Sln: Note that we cannot rewrite the limit as the product of $\lim_{x\to 0} x^2$ and $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$ $\left[\lim_{x\to a}(f\cdot g) = \left(\lim_{x\to a} f(x)\right)\cdot\left(\lim_{x\to a} g(x)\right)\right]$

b/c $\lim_{x\to 0} \sin\left(\frac{1}{x}\right) = DNE$

We know that $-1 \le \sin\theta \le 1$ for all $\theta \in [0, 2\pi]$.

So, in particular $-1 \le \sin\left(\frac{1}{x}\right) \le 1$, since $x^2 \ge 0$ we can multiply both sides by x^2

$$-1\cdot x^2 \le x^2 \sin\left(\frac{1}{x}\right) \le 1\cdot x^2$$

$$\underbrace{-x^2}_{f(x)} \le \underbrace{x^2 \sin\left(\frac{1}{x}\right)}_{g(x)} \le \underbrace{x^2}_{h(x)}$$

Besides, $\lim_{x\to 0} f(x) = \lim_{x\to 0} (-x^2) = 0$ and $\lim_{x\to 0} h(x) = \lim_{x\to 0} (x^2) = 0$

So, $\lim_{x\to 0} f(x) = \lim_{x\to 0} h(x) = 0$ By the squeeze theorem we

$$\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$

Example. Compute $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x^2}\right)$.

Sln: We cannot apply the product law.

We know that $-1 \le \cos\left(\frac{1}{x^2}\right) \le 1$ for all $x \in \mathbb{R}\setminus\{0\}$.

Since $x^2 \ge 0$ then

$$\underbrace{-x^2}_{f(x)} \le \underbrace{x^2 \cos\left(\frac{1}{x^2}\right)}_{g(x)} \le \underbrace{x^2}_{h(x)}$$

Besides, $\lim_{x\to 0} f(x) = 0$ and $\lim_{x\to 0} h(x) = 0$ then since $\lim_{x\to 0} f(x) = \lim_{x\to 0} h(x)$

then By Squeeze theorem $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x^2}\right) = 0$

Some Problems. 1. $\lim_{h\to 0} \dfrac{(-2+h)^{-1}+2^{-1}}{h}$; $x^{-n} = \dfrac{1}{x^n}$

Sln.

$$\frac{(-2+h)^{-1}+2^{-1}}{h} = \frac{\frac{1}{(-2+h)}+\frac{1}{2}}{h} = \frac{\frac{2+(-2+h)}{2(-2+h)}}{\frac{h}{1}} = \frac{2+(-2+h)}{2h(-2+h)}$$

$$= \frac{\cancel{h}}{2\cancel{h}(-2+h)} = \frac{1}{2(-2+h)}$$

$$\lim_{h\to 0} \frac{(-2+h)^{-1}+2^{-1}}{h} = \lim_{h\to 0} \frac{1}{2(-2+h)} = -\frac{1}{4}.$$

2. $\lim_{t\to 0} \left(\frac{1}{t} - \frac{1}{t^2+t}\right)$ $\left(\lim_{x\to a}(f(x)-g(x)) = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)\right)$

Sln:

$$\frac{1}{t} - \frac{1}{t^2+t} = \frac{t^2+t-t}{t(t^2+t)} = \frac{t^2}{t(t(t+1))} = \frac{t^2}{t^2(t+1)} = \frac{1}{t+1}.$$

So, $\lim_{t\to 0} \left(\frac{1}{t} - \frac{1}{t^2+t}\right) = \lim_{t\to 0} \left(\frac{1}{t+1}\right) = \frac{1}{0+1} = 1.$

Continuity of a function

Sometimes we can compute $\lim_{x\to a} f(x) = f(a)$ (by Direct Substitution)
Functions having this property are called continuous at $x=a$.

Def. A function f is continuous at $x=a$ if

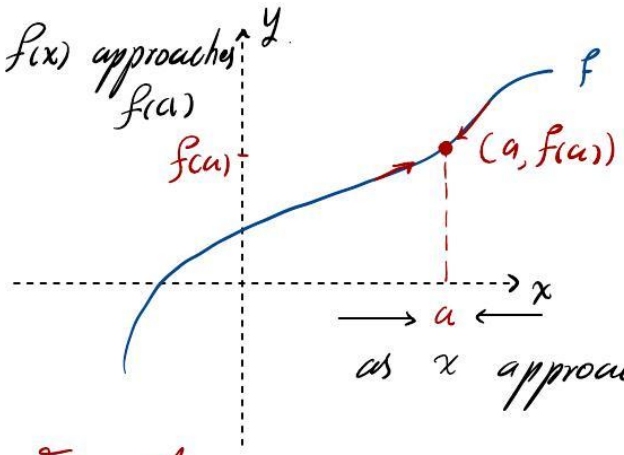
$$\lim_{x\to a} f(x) = f(a).$$

Rmk. Note that this Definition implicitly requires three things, if f is continuous at a.

i). $f(a)$ is defined ($a \in \mathrm{Dom} f$).

ii). $\lim_{x\to a} f(x)$ exists (i.e, $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$)

iii). $\lim_{x\to a} f(x) = f(a)$.

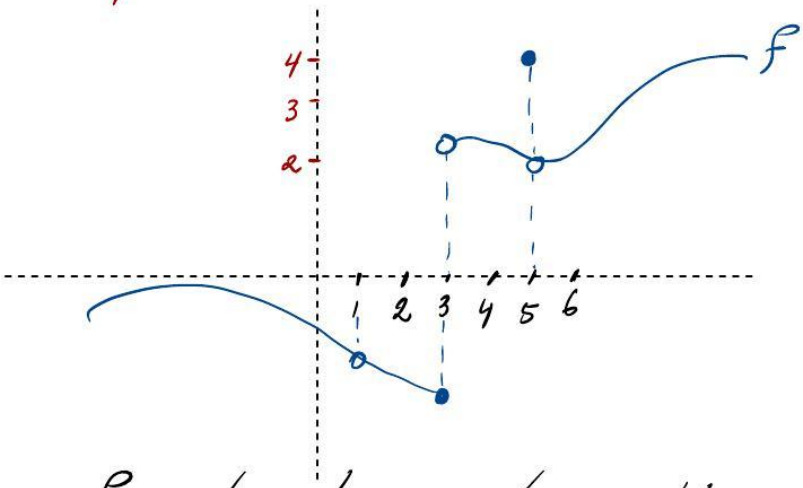


as x approaches a.

Notice that the function f is not continuous at $x=1, 3, 5$.

- f is not continuous at $x=1$ $f(1)$ is not defined (i).
- f is not continuous at $x=3$. b/c $\lim_{x\to 3^-} f(x) \neq \lim_{x\to 3^+} f(x)$ then the limit DNE (ii).

Example



- f is not continuous at $x=5$ b/c $\lim_{x\to 5} f(x) \neq f(5)$.

" if f is not continuous at a then f is discontinuous at a ".

Example. Where are each of the following function discontinuous?

a). $f(x) = \dfrac{x^2-x-2}{x-2}$

b). $f(x) = \begin{cases} \dfrac{x^2-x-2}{x-2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$

c). $f(x) = \begin{cases} \dfrac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$

Sln:

a) Notice that $f(2)$ is not defined, so f is discontinuous at $x=2$.

b) Notice that $f(2)$ is defined, $f(2)=1$.

$$\lim_{x\to 2} f(x) = \lim_{x\to 2} \frac{x^2-x-2}{x-2} = \lim_{x\to 2} \frac{(x-2)(x+1)}{(x-2)} = \lim_{x\to 2} (x+1) = 3.$$

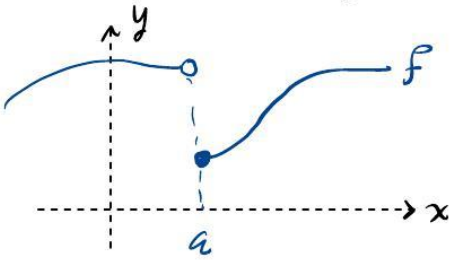
then $\lim_{x\to 2} f(x) \neq f(2)$ then f is discontinuous at $x=2$.

c). Here $f(0)=1$ is defined.

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{1}{x^2} = \infty$$ then f is discontinuous at $x=0$.

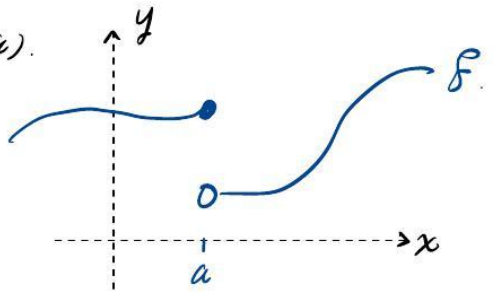
Def. A function f is continuous from the right at a number a if

$$\lim_{x\to a^+} f(x) = f(a)$$



and f is continuous from the left at a number a if

$$\lim_{x\to a^-} f(x) = f(a).$$



Def. A function f is continuous on an interval if it is continuous at every number in the interval.

Example. Show that the function $f(x) = 1 - \sqrt{1-x^2}$ is continuous on the interval $[-1, 1]$.

Sln: If $-1 < a < 1$

$$\lim_{x\to a} f(x) = \lim_{x\to a} \left(1 - \sqrt{1-x^2}\right)$$

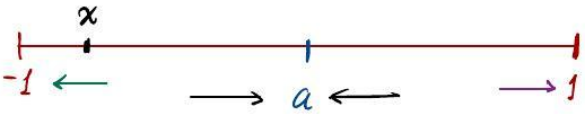
$$= \lim_{x\to a}(1) - \lim_{x\to a}\sqrt{1-x^2}$$

$$= \underbrace{1}_{\lim_{x\to a}(1)} - \sqrt{\lim_{x\to a}(1-x^2)}$$ the root rule $\lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$

$$= 1 - \sqrt{1-a^2}$$ b/c $1-x^2$ is a polynomial

$$= f(a).$$

By definition $f(x)$ is continuous at a, $-1 < a < 1$



$f(-1) = 1$

$$\lim_{x\to -1^+} f(x) = \lim_{x\to -1^+} \left(1 - \sqrt{1-x^2}\right)$$

$$= 1 - \lim_{x\to -1^+} \sqrt{1-x^2}$$

$$= 1 - \sqrt{\lim_{x\to -1^+} (1-x^2)} = 1 - \sqrt{1-1} = 1.$$

Since $f(-1) = \lim_{x\to -1^+} f(x)$ it's continuous at $x=-1$.

For you, similarly $f(1) = 1 = \lim_{x\to 1^-} f(x)$ it's continuous at $x=1$.

Properties of continuous functions.

Thm. Let f, g are continuous functions at a and $c \in \mathbb{R}$. Then, the following are also continuous functions at a.

1. $f+g$ 2. $f-g$ 3. cf 4. $f\cdot g$ 5. $\frac{f}{g}$ if $g(a) \neq 0$.

Thm. a). Any polynomial is continuous everywhere, that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$.

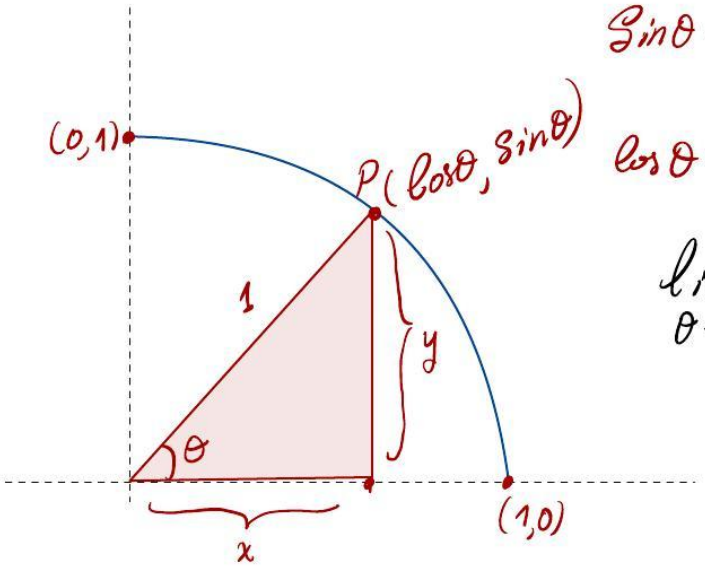
b) Any rational function is continuous whenever it is defined, that is, it is continuous on its domain.

Example. Find $\lim_{x\to -2} \dfrac{x^3+2x^2-1}{5-3x}$

Sln: Let $f(x) = \dfrac{x^3+2x^2-1}{5-3x}$, is rational then by thm the function is continuous on its domain.

We need compute the domain $5-3x=0 \Rightarrow x = \frac{5}{3}$ then $\text{Dom} f = \mathbb{R} \setminus \{\frac{5}{3}\}$ Notice that $-2 \in \text{Dom} f$ so.

$$\lim_{x\to -2} \frac{x^3+2x^2-1}{5-3x} = f(-2) = \frac{(-2)^3+2(-2)-1}{5-3(-2)} = -\frac{1}{11}$$



$\sin\theta = \frac{y}{1} \Rightarrow y = \sin\theta$.

$\cos\theta = \frac{x}{1} \Rightarrow x = \cos\theta$

$\lim_{\theta\to 0} \cos\theta = 1$ and $\lim_{\theta\to 0} \sin\theta = 0$.

We know that $\tan x = \frac{\sin x}{\cos x}$ is continuous except where $\cos x = 0$.
this happens when x is an odd integer multiple of $\frac{\pi}{2}$ so
$y = \tan x$ has infinite discontinuities when $x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \dots$

Thm The following types of functions are continuous at every number in their domains.

- Polynomials ($\mathrm{Dom} = \mathbb{R}$)
- Rational functions ($\mathrm{Dom}\left(\frac{f}{g}\right) = (\mathrm{Dom} f \cap \mathrm{Dom} g) \setminus \{g = 0\}$)
- root functions $\sqrt[n]{f(x)}$
  - n is even: $\mathrm{Dom}\sqrt[n]{f} = \mathrm{Dom} f \cap \{f(x) \geq 0\}$
  - n is odd: $\mathrm{Dom}\sqrt[n]{f} = \mathrm{Dom} f$.
- trigonometric functions, $\mathrm{Dom}\sin x = \mathbb{R}$, $\mathrm{Dom}\cos x = \mathbb{R}$.

Example On what intervals is each function continuous?

a). $f(x) = x^{100} - 2x^{37} + 75$.

Sln. f is a polynomial, so it's continuous on $(-\infty, \infty)$.

b). $g(x) = \dfrac{x^2+2x+17}{x^2-1}$

8h.

c). $h(x) = \sqrt{x} + \dfrac{x+1}{x-1} - \dfrac{x+1}{x^2+1}$